

Trusted Platform Module

- The Trusted Platform Module (TPM) is a low-cost security module that delivers the foundation of a safe computing environment.
- Implemented as a tamper-resistant integrated circuit, it's designed as a building block for trusted computing and is bound to a specific platform.
- TPM allows users to assess the trustworthiness of computers they interact with and creates a foundation of trust for software processes.
- The Trusted Computing Group (TCG) defines trusted computing as the expectation that "a device will behave in a particular manner for a specific purpose."



Protected Capabilities

Secure access to memory and registers



Integrity Measurement

Obtaining platform metrics



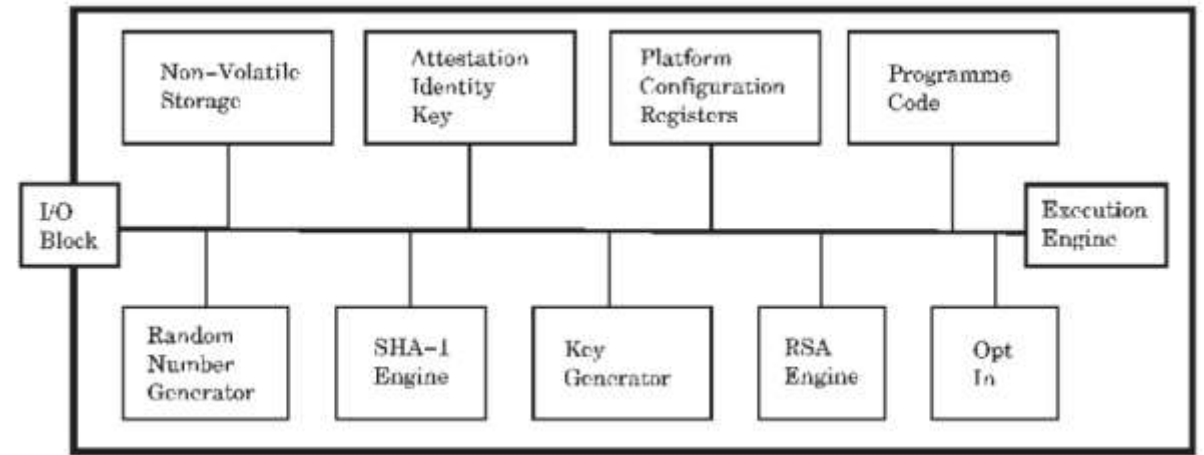
Storage & Reporting

Storing and presenting results

TPM in Modern Applications

- Self-Encrypting Drives
 - Hardware-based data security encrypting data as written to drive, using Media Encryption Key (MEK) and Key Encryption Key (KEK) for locking and unlocking.
- Windows 8 Secure Boot
 - Measures BIOS and reports remotely to enterprise networks, certifying no malware has changed the BIOS since last network connection.
- Mobile Security
 - Windows 8 phones utilize TPM to protect device and user identity, enabling remote device disabling and data erasure if needed.

TPM Architecture



- I/O Block
 - Manages information flow between components via bus and enforces access policies
- Non-Volatile Storage
 - Stores long-term keys including Endorsement Key (EK) and Storage Root Key (SRK)
- Cryptographic Engines
 - Houses SHA-1 engine, RSA key generation, and RSA encryption/decryption capabilities
- Platform Configuration Registers
 - Stores integrity metrics measuring code integrity from BIOS to applications

Endorsement Key: The Foundation of Trust

- The Endorsement Key (EK) is a unique feature embedded in each TPM before shipping.
 - The private key is permanently embedded and **unique to each TPM**, while the public key is stored in a certificate.
- EK is used to generate Attestation Identity Keys (AIKs) for routine operations and encrypting data during the ownership process.
 - A certificate signed by a Trusted Platform Module Entity (TPME) verifies the public key corresponds to a private key stored in a genuine TPM.

Private Key

Permanently embedded, unique to each TPM platform

Public Key

Stored in certificate, used for limited operations

AIK Generation

Creates aliases for routine operations, enabling user anonymity

Platform Configuration Registers (PCRs)

■ Purpose

- Store integrity metrics measuring code integrity before launching, from BIOS to applications.

■ Requirements

- At least 16 PCRs of 20 bytes each.
- Must reset at system restart or power loss.

■ Allocation

- Registers 0-7 reserved for TPM use.
- Registers 8-15 free for OS and applications.

How TPM Protects Your Data



Key Storage

Sensitive data like cryptographic keys stored in secure hardware location, never leaving TPM



Hardware Authentication

TPM chip uniquely identifies hardware, preventing key theft and reuse attacks



Restricted Operations

Only safe actions allowed - can encrypt with key but cannot access key directly

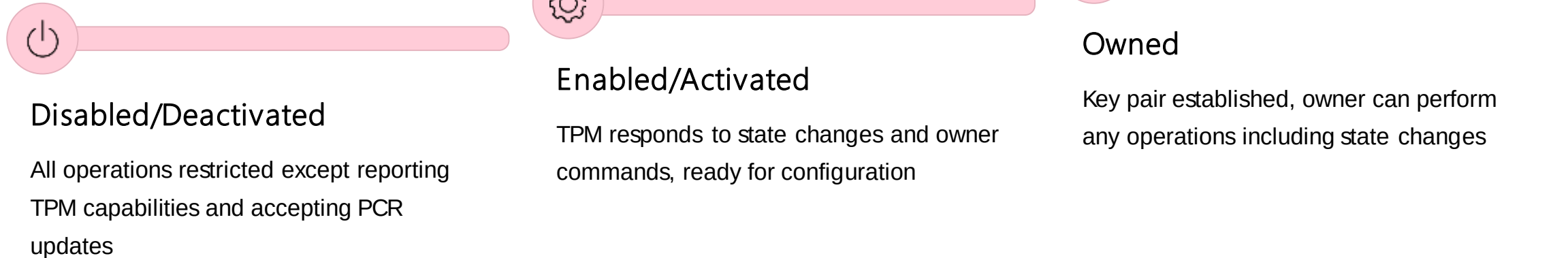
Traditional Security Flaw

Existing security systems run on unknown hardware. If one system is compromised, private keys can be copied and used for attacks later. File-based security approaches are fundamentally vulnerable.

TPM Advantage

Even with unrestricted computer access, attackers cannot access sensitive data stored on TPM. Similar to mobile phone authentication, TPM authenticates the hardware itself.

TPM Operational States



 **Opt-In Requirement:** Users must take ownership and configure the TPM. Changing state flags requires authorization, with the opt-in block providing mechanisms to maintain TPM state protection.